

# Validated Isolation of a Negative Continuum Resonance for a Folded-Gaussian Markov Operator

A Grushin–Rouché Certificate and an Exact Dyadic Galerkin Lift

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## Abstract

A preceding weighted-Riesz analysis reduced the negative peripheral branch of a folded-Gaussian Markov family to one explicit hypothesis: the full continuum operator must possess a simple isolated negative resonance, with enough contour-resolvent control to survive discretization. Convergent finite-grid eigenvalues do not prove that statement for a nonnormal operator. We close the full-kernel continuum gate at the fixed noise width  $\sigma = 10^{-2}$  and at the first band-merging parameter

$$u_c^3 - 2u_c^2 + 2u_c - 2 = 0.$$

Let  $\mathcal{K}$  be the exactly normalized folded-Gaussian Markov operator on  $L^\infty([0, 1])$  generated by  $m(x) = 1 - u_c x^2$ . For

$$c = -0.9865481927458079, \quad r = 0.05, \quad \Gamma = \{z \in \mathbb{C} : |z - c| = r\},$$

we prove that  $\Gamma$  belongs to the resolvent set of  $\mathcal{K}$ , that its interior contains exactly one eigenvalue counted with algebraic multiplicity, and that

$$\sup_{z \in \Gamma} \|(z - \mathcal{K})^{-1}\|_{L^\infty \rightarrow L^\infty} \leq 148.84153451786617.$$

The operator and the circle are conjugation invariant and the circle lies strictly in the negative half-plane. The enclosed eigenvalue is therefore real, negative, and algebraically simple.

The proof has four independently audited layers. First, a balanced rank-one Grushin problem at dimension 4096 is validated against the exact stored binary64 Markov matrix. A componentwise outward residual of  $4.588 \times 10^{-11}$ , a one-center transport product of 0.499527, and a scalar Rouché error of  $3.278 \times 10^{-13}$  prove one enclosed matrix eigenvalue and a contour-resolvent upper of 41.899. Second, 224-bit Arb arithmetic bridges that exact stored matrix to the exact-parameter continuum-normalized midpoint matrix with row- $\ell^1$  defect at most  $7.315 \times 10^{-6}$ . Third, cell-average Galerkin matrices satisfy exact coarse consistency and explicit  $L^\infty$  Haar bounds, allowing two Schur continuations  $4096 \rightarrow 8192 \rightarrow 16384$ . Fourth, a finite-rank complement estimate and a continuum operator defect of 0.012476 close the final Neumann gate with product 0.649969.

This validates the simple-isolated full-kernel parity premise of the preceding weighted-Riesz theorem. It also gives a uniform  $L^\infty$  contour theorem for the exact continuum-normalized midpoint family from dimension 32768 onward. The validated norm is not a dimension-uniform Euclidean norm for the archived sparse matrices. No zero-noise limit, zeta-zero identification, self-adjoint Hilbert–Pólya operator, or Riemann-hypothesis claim is made.

## 1 Introduction

The nested-grid program of Wang [10, 11] follows a noisy quadratic transfer operator through dyadic discretizations. Its physical two-step operator removes a Perron contribution and a

negative peripheral contribution before squaring the remainder. A smooth-kernel Haar theorem subsequently explained the observed quarter-half law

$$\|E_h\|, \|D_h\| = O(h^2), \quad \|C_h\|, \|B_h\| = O(h),$$

and a separate Gaussian-tail theorem treated the hard support cutoff [8, 9].

The remaining peripheral issue was spectral rather than algebraic. Wang [13] replaced gauge-dependent left and right eigenvectors by the intrinsic weighted Riesz term

$$\mathcal{Q}(A; \Gamma) = \frac{1}{2\pi i} \int_{\Gamma} z(z - A)^{-1} dz.$$

For a smooth compact operator, a simple isolated nonzero eigenvalue gives a second-order Nyström bridge for this weighted term. Strong positivity makes the Perron application unconditional. The negative branch was left conditional on a simple isolated continuum resonance.

That condition cannot be inferred from a short table of finite-grid eigenvalues. Nonnormal spectra can move much farther than eigenvalue separation suggests, and a perturbation theorem must see the resolvent [4, 7]. In the present problem the missing statement has two parts:

1. a contour must contain exactly one eigenvalue of the full continuum operator, counted algebraically; and
2. the resolvent on that contour must have an explicit norm upper that survives the finite-to-continuum passage.

We prove both statements by combining a one-center Grushin reduction with a conditional-expectation Galerkin hierarchy. The main contributions are:

1. a moment proof of explicit first- and second-derivative row- $L^1$  bounds for the normalized folded-Gaussian kernel;
2. exact coarse consistency for cell-average Galerkin operators and explicit  $L^\infty$  Haar block bounds;
3. a rank-one Grushin–Rouché theorem that turns one validated inverse at the circle center into an exact eigenvalue count and contour resolvent;
4. a 224-bit Arb bridge from the archived sparse binary64 matrix to the exact-parameter continuum midpoint matrix;
5. two dyadic Schur continuations and a final finite-rank-to-continuum Neumann transfer; and
6. a sharp theorem boundary separating the resulting continuum  $L^\infty$  statement from any unproved dimension-uniform Euclidean claim for sparse midpoint matrices.

Three evidence levels are kept separate throughout.

### **Analytic operator theorem**

Moment inequalities, Galerkin identities, Schur bounds, perturbation invariance, and the continuum conclusion.

### **Outward certificate**

Componentwise binary64 residual enclosures and 224-bit Arb evaluations of exact stored and exact-parameter quantities.

### **Floating diagnostic**

The leading-spectrum points used only in the summary figure. They are never used to establish the root count.

## 2 The exact operator and the main theorem

Let

$$p(u) = u^3 - 2u^2 + 2u - 2.$$

Since

$$p'(u) = 3u^2 - 4u + 2 = 3\left(u - \frac{2}{3}\right)^2 + \frac{2}{3} > 0,$$

the polynomial has a unique real root  $u_c$  in  $(1.5, 1.6)$ . Fix

$$\sigma = \frac{1}{100}, \quad m(x) = 1 - u_c x^2, \quad 0 \leq x \leq 1.$$

Because  $1 < u_c < 2$ , one has  $m([0, 1]) = [1 - u_c, 1] \subset [-1, 1]$ . For  $m \in [-1, 1]$ , define

$$g_m(y) = \exp\left[-\frac{(y - m)^2}{2\sigma^2}\right] + \exp\left[-\frac{(y + m)^2}{2\sigma^2}\right], \quad 0 \leq y \leq 1,$$

$$Z(m) = \int_0^1 g_m(y) dy, \quad k(x, y) = \frac{g_{m(x)}(y)}{Z(m(x))}.$$

The continuum Markov operator is

$$(\mathcal{K}f)(x) = \int_0^1 k(x, y) f(y) dy \quad \text{on } L^\infty([0, 1]). \quad (1)$$

Its kernel is real, strictly positive, and smooth on the compact square. Thus  $\mathcal{K}$  is a compact Markov operator; in particular,  $\|\mathcal{K}\|_{\infty \rightarrow \infty} = 1$ .

The numerical contour data are

$$c = -0.9865481927458079, \quad r = 0.05, \quad \Gamma = \{z \in \mathbb{C} : |z - c| = r\}. \quad (2)$$

The certificate treats the round-trip binary64 values represented by the displayed decimals as exact real inputs and rounds every subsequent inequality outward. The contour obeys

$$\min_{z \in \Gamma} |z| \geq 0.9365481927458077, \quad \max_{z \in \Gamma} \operatorname{Re} z = -0.9365481927458078 < 0.$$

**Theorem 2.1** (Validated negative continuum resonance). *For the exact operator (1) at  $\sigma = 1/100$  and the exact critical parameter  $u_c$ :*

(i)  $\Gamma \subset \rho(\mathcal{K})$ ;

(ii) *the Riesz projection*

$$\Pi_\Gamma(\mathcal{K}) = \frac{1}{2\pi i} \int_\Gamma (z - \mathcal{K})^{-1} dz$$

*has rank one;*

(iii) *the interior of  $\Gamma$  contains one real negative algebraically simple eigenvalue  $\lambda_-$ ; and*

(iv)

$$\sup_{z \in \Gamma} \|(z - \mathcal{K})^{-1}\|_{\infty \rightarrow \infty} \leq 148.84153451786617.$$

The proof is completed in [section 7](#). The reality conclusion does not require a separate numerical enclosure of the imaginary part: a real operator has conjugate-paired spectrum, while a conjugation-invariant disk of algebraic count one cannot contain a nonreal pair.

**Corollary 2.2** (Closure of the full-kernel parity premise). *At  $\sigma = 1/100$ , the simple-isolated negative-resonance premise in the full-kernel parity corollary of Wang [13] is satisfied. Hence the gauge-free weighted Nyström term for this branch has the second-order full-kernel continuum bridge proved there.*

*Remark 2.3* (Norm boundary). **Theorem 2.1** is an  $L^\infty$  operator theorem. It does not assert a dimension-uniform  $\ell^2$  resolvent bound for the support-truncated stored matrices. This distinction is essential for a nonnormal family.

### 3 Moment and derivative bounds for the folded kernel

The continuum step requires explicit row-integrated derivative bounds rather than pointwise Gaussian peaks. They follow from a truncated-normal moment inequality.

Let  $T$  have density

$$\pi_m(t) = \frac{\exp[-(t-m)^2/(2\sigma^2)]}{\int_{-1}^1 \exp[-(s-m)^2/(2\sigma^2)] ds}, \quad -1 \leq t \leq 1,$$

where  $m \in [-1, 1]$ . Folding  $T$  by  $t \mapsto |t|$  gives the density  $g_m/Z(m)$  on  $[0, 1]$ .

**Lemma 3.1** (Truncated second moment). *For every  $m \in [-1, 1]$ ,*

$$\mathbb{E}(T - m)^2 \leq \sigma^2, \quad \text{Var}(T) \leq \sigma^2.$$

*Proof.* Put  $X = (T - m)/\sigma$ . Its conditioning interval is  $[a, b]$  with  $a \leq 0 \leq b$ . Since

$$\int_a^b (x^2 - 1)e^{-x^2/2} dx = [-xe^{-x^2/2}]_a^b = -be^{-b^2/2} + ae^{-a^2/2} \leq 0,$$

the conditional expectation satisfies  $\mathbb{E}X^2 \leq 1$ . This proves the first inequality, and variance is no larger than the second moment about  $m$ .  $\square$

**Lemma 3.2** (Normalized folded derivative envelope). *Write  $q_m(y) = g_m(y)/Z(m)$ . Then*

$$\|\partial_m q_m\|_{L^1(0,1)}, \quad \|\partial_y q_m\|_{L^1(0,1)} \leq \sigma^{-1}, \quad (3)$$

$$\|\partial_{mm} q_m\|_{L^1(0,1)}, \quad \|\partial_{my} q_m\|_{L^1(0,1)}, \quad \|\partial_{yy} q_m\|_{L^1(0,1)} \leq 2\sigma^{-2}. \quad (4)$$

*Proof.* Let  $\mu = \mathbb{E}T$  and  $v = \text{Var}(T)$ . Before folding, the parameter score identities are

$$\begin{aligned} \partial_m \pi_m(t) &= \frac{t - \mu}{\sigma^2} \pi_m(t), \\ \partial_{mm} \pi_m(t) &= \frac{(t - \mu)^2 - v}{\sigma^4} \pi_m(t). \end{aligned}$$

By lemma 3.1, Cauchy–Schwarz gives the first parameter bound and

$$\|\partial_{mm} \pi_m\|_1 \leq \frac{2v}{\sigma^4} \leq \frac{2}{\sigma^2}.$$

The target identities are

$$\partial_t \pi_m(t) = -\frac{t - m}{\sigma^2} \pi_m(t), \quad \partial_{tt} \pi_m(t) = \left( \frac{(t - m)^2}{\sigma^4} - \frac{1}{\sigma^2} \right) \pi_m(t),$$

and

$$\partial_{mt} \pi_m(t) = \left( \frac{1}{\sigma^2} - \frac{(t - m)(t - \mu)}{\sigma^4} \right) \pi_m(t).$$

Again lemma 3.1 and Cauchy–Schwarz give the displayed bounds. Pushforward by the folding map cannot increase total variation, so the same  $L^1$  inequalities hold for  $q_m$ .  $\square$

For  $m(x) = 1 - u_c x^2$ , use the validated analytic upper  $u_c < 1.544$ . Define

$$L_x = \sup_x \int_0^1 |\partial_x k(x, y)| dy, \quad L_y = \sup_x \int_0^1 |\partial_y k(x, y)| dy,$$

and analogously  $L_{xx}, L_{xy}, L_{yy}$ .

**Corollary 3.3** (Numerical derivative ledger). *The exact continuum kernel satisfies the outward bounds*

$$\begin{aligned} L_x &\leq 308.80000000000024, & L_y &\leq 100.00000000000001, \\ L_{xx} &\leq 191023.680000000023, & L_{xy} &\leq 61760.0000000000044, \\ L_{yy} &\leq 20000.0000000000004. \end{aligned}$$

*Proof.* The chain rule and [lemma 3.2](#) give

$$\begin{aligned} L_x &\leq 2u\sigma^{-1}, & L_{xx} &\leq 4u^2(2\sigma^{-2}) + 2u\sigma^{-1}, \\ L_{xy} &\leq 2u(2\sigma^{-2}), & L_y &\leq \sigma^{-1}, & L_{yy} &\leq 2\sigma^{-2}. \end{aligned}$$

Insert  $u = 1.544$  and  $\sigma = 1/100$ , rounding upward.  $\square$

## 4 Cell-average Galerkin and contour transport

Let  $I_i = [ih, (i+1)h)$ ,  $h = 1/n$ , and let  $P_n$  be conditional expectation onto functions constant on the cells  $I_i$ . The finite-rank operator

$$\mathcal{A}_n = P_n \mathcal{K} P_n$$

is represented on cell values by the row-stochastic matrix

$$(A_n)_{ij} = \frac{1}{h} \int_{I_i} \int_{I_j} k(x, y) dy dx. \quad (5)$$

The continuum-normalized midpoint matrix is

$$(M_n)_{ij} = h k(x_i, y_j), \quad x_i = y_i = \left(i + \frac{1}{2}\right) h. \quad (6)$$

All matrix norms in this paper are induced row- $\ell^\infty$  norms unless another norm is explicitly written.

**Proposition 4.1** (Midpoint and continuum defects). *The derivative constants of [corollary 3.3](#) give*

$$\|M_n - A_n\|_\infty \leq \frac{h^2}{8} (L_{xx} + L_{yy}), \quad (7)$$

$$\|\mathcal{K} - P_n \mathcal{K} P_n\|_{\infty \rightarrow \infty} \leq \frac{h}{2} (L_x + L_y). \quad (8)$$

*Proof.* For (7), apply the one-dimensional midpoint remainder successively in  $x$  and  $y$  on every cell rectangle. The entrywise error is bounded by  $h^3(L_{xx} + L_{yy})/8$ ; summing one matrix row over  $n = 1/h$  cells gives the claim.

For (8), write

$$\mathcal{K} - P_n \mathcal{K} P_n = (\mathbf{I} - P_n) \mathcal{K} + P_n \mathcal{K} (\mathbf{I} - P_n).$$

Output averaging costs at most  $hL_x/2$ . On each input cell,  $f - P_n f$  has mean zero, so subtracting a cellwise constant from  $k(x, \cdot)$  and using the target derivative costs at most  $hL_y/2$ .  $\square$

The advantage of cell averages over midpoint samples is exact nesting. Let  $J, W : \mathbb{C}^n \rightarrow \mathbb{C}^{2n}$  be replication and alternating-detail injections,

$$(Jv)_{2i} = (Jv)_{2i+1} = v_i, \quad (Wv)_{2i} = v_i, \quad (Wv)_{2i+1} = -v_i,$$

with restrictions  $R = J^*/2$  and  $S = W^*/2$ . In these coordinates,

$$\begin{bmatrix} R \\ S \end{bmatrix} A_{2n} [J \ W] = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix}. \quad (9)$$

The upper-left identity is exact because  $P_n P_{2n} = P_n$ .

**Proposition 4.2** (Exact-consistency Haar bounds). *For a coarse mesh  $h = 1/n$ ,*

$$\|C_n\|_\infty \leq \frac{h}{4} L_x, \quad \|B_n\|_\infty \leq \frac{h}{2} L_y, \quad \|D_n\|_\infty \leq \frac{h^2}{8} L_{xy}. \quad (10)$$

*The coarse-consistency defect in (9) is exactly zero.*

*Proof.* The  $C_n$  block contains one signed difference between the two source half-cells, the  $B_n$  block one mean-zero target-cell cancellation, and the  $D_n$  block both cancellations. Integrating the corresponding first or mixed derivative over the half-cell rectangles gives (10). Exact coarse consistency follows from the tower property of conditional expectation.  $\square$

We use two standard contour-transfer lemmas. They are stated explicitly to keep every nonnormal constant visible.

**Lemma 4.3** (Neumann contour transfer). *Let  $\Gamma \subset \rho(A)$  and*

$$M = \sup_{z \in \Gamma} \|(z - A)^{-1}\|.$$

*If  $\|B - A\| \leq \delta$  and  $M\delta < 1$ , then  $\Gamma \subset \rho(B)$ ,*

$$\sup_{z \in \Gamma} \|(z - B)^{-1}\| \leq \frac{M}{1 - M\delta},$$

*and the Riesz projections of  $A$  and  $B$  inside  $\Gamma$  have the same finite rank.*

*Proof.* Factor  $z - B = (z - A)[I - (z - A)^{-1}(B - A)]$  and use a Neumann series uniformly on the contour. The same estimate applies to the homotopy  $A + t(B - A)$ ,  $0 \leq t \leq 1$ , so the Riesz rank is constant [4].  $\square$

**Lemma 4.4** (One dyadic Schur step). *Suppose the coarse contour resolvent is bounded by  $M$ , and put*

$$\rho = \min_{z \in \Gamma} |z|, \quad b = \|B_n\|_\infty, \quad c_1 = \|C_n\|_\infty, \quad d = \|D_n\|_\infty.$$

*If  $d < \rho$ , set*

$$R_d = \frac{1}{\rho - d}, \quad \varepsilon = b R_d c_1.$$

*If  $M\varepsilon < 1$ , the fine and coarse matrices have the same count inside  $\Gamma$ . Moreover, with*

$$S = \frac{M}{1 - M\varepsilon},$$

*the fine resolvent is bounded by*

$$S + SbR_d + R_d c_1 S + R_d + R_d c_1 S b R_d. \quad (11)$$

*Proof.* Take the Schur complement of  $z - D_n$ . Its inverse is bounded by  $R_d$ , and its self-energy has norm at most  $\varepsilon$ . Apply [lemma 4.3](#) to the coarse Schur complement and then bound the four blocks of the inverse. For count preservation, use the homotopy

$$\begin{pmatrix} A_n & tB_n \\ tC_n & tD_n \end{pmatrix}, \quad 0 \leq t \leq 1.$$

Every displayed bound is dominated by its value at  $t = 1$ , so the contour remains in the resolvent set throughout. Formula (11) also includes the exact conversion from coarse/detail coordinates back to the fine  $\ell^\infty$  coordinates.  $\square$

**Lemma 4.5** (The zero complement). *If the Galerkin matrix  $A_n$  has contour-resolvent upper  $M_n$ , then the finite-rank operator  $\mathcal{A}_n = P_n \mathcal{K} P_n$  on all of  $L^\infty$  satisfies*

$$\sup_{z \in \Gamma} \|(z - \mathcal{A}_n)^{-1}\|_{\infty \rightarrow \infty} \leq M_n + \frac{2}{\rho}, \quad \rho = \min_{z \in \Gamma} |z|. \quad (12)$$

*Proof.* On  $\text{Ran } P_n$ , the operator is represented by  $A_n$ ; on  $\ker P_n$  it is zero. Therefore

$$(z - \mathcal{A}_n)^{-1} = J_n(z - A_n)^{-1} P_n + z^{-1}(\mathbf{I} - P_n),$$

where  $J_n$  includes cell values as piecewise-constant functions. Since  $\|P_n\| = 1$  and  $\|\mathbf{I} - P_n\| \leq 2$ , (12) follows.  $\square$

## 5 The exact stored Grushin certificate

Let  $P_{4096}^s$  denote the sparse Markov matrix archived in the RH-36 snapshot. In this section every stored binary64 entry is treated as an exact real number. The second stored peripheral mode supplies a center  $c$ , right vector  $r_0$ , and left vector  $\ell_0$ . No assumption that these are exact eigenvectors is made.

With the exactly representable scale  $s = 16$ , define the bordered problem

$$\mathcal{G}(z) = \begin{pmatrix} z\mathbf{I} - P_{4096}^s & sr_0 \\ s^{-1}\ell_0^T & 0 \end{pmatrix}. \quad (13)$$

Write its inverse blocks as

$$\mathcal{G}(z)^{-1} = \begin{pmatrix} E(z) & E_+(z) \\ E_-(z) & E_{-+}(z) \end{pmatrix}.$$

**Proposition 5.1** (One-center Grushin–Rouché criterion). *Let*

$$g_- \leq \ell_0^T r_0 \leq g_+, \quad \delta_r = \|(P_{4096}^s - c\mathbf{I})r_0\|_\infty, \\ \delta_\ell = \|((P_{4096}^s)^T - c\mathbf{I})\ell_0\|_1, \quad e_0 \geq \|E(c)\|_\infty.$$

*If  $re_0 < 1$ , put*

$$M_E = \frac{e_0}{1 - re_0}, \\ M_+ = \frac{s\|r_0\|_\infty}{g_-} + \frac{M_E s \delta_r}{g_-}, \quad M_- = \frac{\|\ell_0\|_1}{sg_-} + \frac{\delta_\ell M_E}{sg_-}, \\ \varepsilon_* = \frac{M_- s \delta_r}{g_-}.$$

*If*

$$\varepsilon_* < \frac{r}{g_+},$$

*then the disk  $|z - c| < r$  contains exactly one eigenvalue of  $P_{4096}^s$  counted algebraically, and*

$$\sup_{|z-c|=r} \|(z - P_{4096}^s)^{-1}\|_\infty \leq M_E + \frac{M_+ M_-}{r/g_+ - \varepsilon_*}. \quad (14)$$

*Proof.* The bordered resolvent identity gives

$$E(z) = E(c)[I + (z - c)E(c)]^{-1}.$$

Thus  $re_0 < 1$  keeps  $\mathcal{G}(z)$  invertible on the closed disk and bounds the leading block by  $M_E$ . Applying the exact bordered inverse identities to  $r_0$  and  $\ell_0$  gives the channel bounds  $M_+$  and  $M_-$ . Applying  $\mathcal{G}(z)^{-1}$  to the candidate border column

$$\begin{pmatrix} sr_0/g \\ -(z - c)/g \end{pmatrix}, \quad g = \ell_0^T r_0,$$

shows that on the boundary

$$\left| E_{-+}(z) + \frac{z - c}{g} \right| \leq \varepsilon_*.$$

Condition (5.1) and scalar Rouché imply that  $E_{-+}$  has one zero in the disk [1]. Since the bordered problem has no pole there, zeros of the effective scalar are eigenvalues of the original matrix with algebraic multiplicity [2, 6]. Finally,

$$(z - P_{4096}^s)^{-1} = E(z) - E_+(z)E_{-+}(z)^{-1}E_-(z),$$

which gives (14). □

The approximate inverse of  $\mathcal{G}(c)$  is generated by sparse LU. Its residual is then recomputed against the exact stored graph

$$cX - P_{4096}^s X + sr_0\alpha, \quad s^{-1}\ell_0^T X,$$

using componentwise outward arithmetic [5, 12]. In particular, the rounded sparse assembly used to generate the LU factor is not reused as the exact target.

Table 1: Exact-stored 4096 Grushin ledger. All displayed inequalities are outward upper or lower bounds.

| quantity                                                   | certified value                            |
|------------------------------------------------------------|--------------------------------------------|
| bordered inverse residual $\ I - \mathcal{G}(c)X\ _\infty$ | $4.587902979431811 \times 10^{-11}$        |
| $\ \mathcal{G}(c)^{-1}\ _\infty$                           | $\leq 10.263377331567700$                  |
| center reduced inverse $e_0$                               | $\leq 9.990523884439735$                   |
| center transport product $re_0$                            | $\leq 0.499526194221987$                   |
| right mode residual $\delta_r$                             | $\leq 5.098820251133382 \times 10^{-15}$   |
| left mode residual $\delta_\ell$                           | $\leq 3.817629187162489 \times 10^{-11}$   |
| Gram interval $\ell_0^T r_0$                               | $[1.0000000000000002, 1.0000000000000024]$ |
| effective-scalar error $\varepsilon_*$                     | $\leq 3.277264606378967 \times 10^{-13}$   |
| effective-scalar boundary lower                            | $\geq 0.04999999999967214$                 |
| stored contour resolvent                                   | $\leq 41.89853481734383$                   |

**Corollary 5.2** (Stored count). *The exact stored binary64 matrix  $P_{4096}^s$  has exactly one eigenvalue in the disk (2), counted algebraically, and its contour resolvent is bounded by 41.89853481734383.*

## 6 Arb bridge to the exact-parameter midpoint matrix

The stored matrix uses a binary64 approximation of  $u_c$ , a support window, binary64 Gaussian values, and discrete row normalization. To connect corollary 5.2 to the analytic operator, all of these effects are enclosed in one row- $\ell^1$  comparison with the exact matrix (6).



At 224-bit precision, Arb [3] validates

$$\begin{aligned} 1.543689012692076361570855971801747986525203297650983935240803 &< u_c \\ &< 1.543689012692076361570855971801747986525203297650983935240805. \end{aligned}$$

The polynomial balls at the endpoints are respectively

$$[-3.08669 \times 10^{-60} +/ - 3.14 \times 10^{-66}]$$

and

$$[2.861655 \times 10^{-60} +/ - 6.08 \times 10^{-67}].$$

Together with  $p' > 0$ , this proves the exact root bracket.

The archived support has half-width 330 midpoint cells. Arb verifies its contiguous support geometry in every one of the 4096 rows throughout the root interval. The smallest distance of a folded center from an ambiguous floor boundary is

$$9.718324434970780 \times 10^{-6}.$$

A Gaussian sum–integral tail estimate gives the uniform omitted exact midpoint mass

$$\tau_{\text{omit}} \leq 1.9049777585621693 \times 10^{-15}.$$

Every retained stored binary64 entry is then subtracted from the exact Arb ball

$$h \frac{e^{-(y_j - m(x_i))^2 / (2\sigma^2)} + e^{-(-y_j - m(x_i))^2 / (2\sigma^2)}}{Z(m(x_i))}$$

and the absolute balls are summed rowwise.

**Proposition 6.1** (Exact stored-to-midpoint bridge). *For the exact stored binary64 matrix and the exact-parameter continuum-normalized midpoint matrix,*

$$\|P_{4096}^s - M_{4096}\|_{\infty} \leq 7.314948554132701 \times 10^{-6}. \quad (15)$$

*The largest enclosed row is row 302.*

This is an Arb statement about exact stored numbers and exact analytic parameters. It is not a floating residual estimate.

Combining [equation \(15\)](#) with [equation \(7\)](#) gives

$$\begin{aligned} \|P_{4096}^s - A_{4096}\|_{\infty} &\leq 7.314948554132701 \times 10^{-6} + 0.0015722489356994657 \\ &= 0.0015795638842535985. \end{aligned} \quad (16)$$

The product with the stored contour-resolvent upper is

$$41.89853481734383 \times 0.0015795638842535985 \leq 0.06618141240061828 < 1.$$

Therefore [lemma 4.3](#) proves:

**Corollary 6.2** (Galerkin count at 4096). *The cell-average Galerkin matrix  $A_{4096}$  has one eigenvalue inside  $\Gamma$ , counted algebraically, and*

$$\sup_{z \in \Gamma} \|(z - A_{4096})^{-1}\|_{\infty} \leq 44.86795976620544.$$

## 7 Dyadic continuation and continuum closure

The exact-consistency hierarchy now propagates [corollary 6.2](#) without constructing matrices at dimensions 8192 or 16384. [Table 2](#) lists the analytic block and Schur bounds.

Table 2: Two exact-consistency dyadic Schur steps. Here  $C$  maps coarse input to detail output and  $B$  maps detail input to coarse output.

| step                     | $\ C\ _\infty$           | $\ B\ _\infty$           | $\ D\ _\infty$           | self-energy              | product    | fine resolvent |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|------------|----------------|
| 4096 $\rightarrow$ 8192  | $1.88477 \times 10^{-2}$ | $1.22070 \times 10^{-2}$ | $4.60148 \times 10^{-4}$ | $2.45782 \times 10^{-4}$ | 0.0110278  | 47.95355       |
| 8192 $\rightarrow$ 16384 | $9.42383 \times 10^{-3}$ | $6.10352 \times 10^{-3}$ | $1.15037 \times 10^{-4}$ | $6.14229 \times 10^{-5}$ | 0.00294545 | 49.96374       |

Both Schur products are below one. Hence the algebraic count remains one at both levels. The resulting 16384 Galerkin matrix resolvent satisfies

$$M_{16384} \leq 49.963731139831324. \quad (17)$$

The matrix estimate must still be lifted to the operator on all of  $L^\infty$ . The minimum contour modulus is

$$\rho \geq 0.9365481927458077.$$

By [lemma 4.5](#), the zero action on  $\ker P_{16384}$  adds

$$\frac{2}{\rho} \leq 2.135501424797291,$$

so

$$\sup_{z \in \Gamma} \left\| (z - P_{16384} \mathcal{K} P_{16384})^{-1} \right\|_{\infty \rightarrow \infty} \leq 52.09923256462863. \quad (18)$$

The analytic continuum defect is

$$\| \mathcal{K} - P_{16384} \mathcal{K} P_{16384} \|_{\infty \rightarrow \infty} \leq 0.012475585937500016. \quad (19)$$

The final Neumann product is therefore

$$52.09923256462863 \times 0.012475585937500016 \leq 0.649968453137824 < 1. \quad (20)$$

Applying [lemma 4.3](#) gives

$$\sup_{z \in \Gamma} \left\| (z - \mathcal{K})^{-1} \right\|_{\infty \rightarrow \infty} \leq \frac{52.09923256462863}{1 - 0.649968453137824} \leq 148.84153451786617.$$

*Proof of theorem 2.1.* [Corollary 5.2](#) establishes algebraic count one for the exact stored matrix. [Corollary 6.2](#) transfers that count to  $A_{4096}$ . [Lemma 4.4](#) and [table 2](#) preserve it through dimensions 8192 and 16384. The zero complement adds only the spectral value 0, which lies outside the disk, and [\(20\)](#) transfers the count to  $\mathcal{K}$ . Thus the Riesz rank is one and the displayed continuum resolvent bound holds.

The kernel is real and  $\Gamma$  is invariant under conjugation. If the unique enclosed eigenvalue were nonreal, its conjugate would also be enclosed with the same algebraic multiplicity, contradicting count one. Hence it is real. The disk lies strictly in the negative half-plane, so the eigenvalue is negative. Algebraic count one makes it simple.  $\square$

**Corollary 7.1** (Uniform exact midpoint family). *Let  $\widetilde{M}_n$  be the piecewise-constant lift of the exact continuum-normalized midpoint matrix [\(6\)](#). For every  $n \geq 32768$ , the interior of  $\Gamma$  contains exactly one eigenvalue of  $M_n$ , counted algebraically, and*

$$\sup_{z \in \Gamma} \left\| (z - \widetilde{M}_n)^{-1} \right\|_{\infty \rightarrow \infty} \leq 2192.043145718256.$$

*Proof.* Proposition 4.1 gives

$$\|\mathcal{K} - \widetilde{M}_n\|_{\infty \rightarrow \infty} \leq \frac{L_x + L_y}{2n} + \frac{L_{xx} + L_{yy}}{8n^2}.$$

The right side decreases with  $n$ . At  $n = 32768$  its outward upper is 0.006262359358370314, whose product with the continuum resolvent upper is  $0.9320991766021576 < 1$ . Lemma 4.3 gives the uniform bound at the threshold, and all larger dimensions have a smaller perturbation.  $\square$

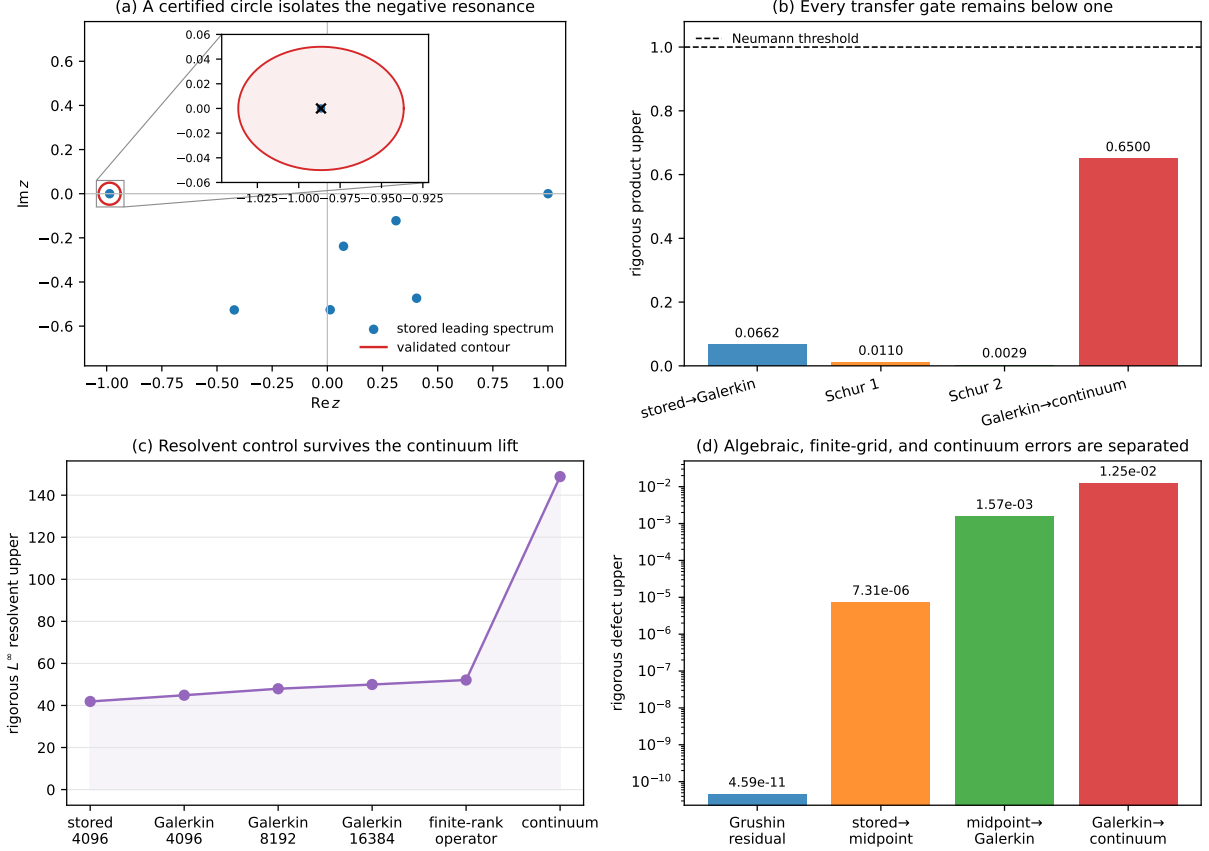


Figure 1: The validated continuum lift. (a) The blue leading-spectrum points are a floating diagnostic from the preceding paper; the red circle is the rigorous contour. (b) Every rigorous Neumann or Schur transfer product is below one. (c) The  $L^\infty$  resolvent upper remains controlled through the stored, Galerkin, finite-rank, and continuum levels. The final jump includes the conservative continuum perturbation. (d) Exact algebraic, stored-to-midpoint, midpoint-to-Galerkin, and Galerkin-to-continuum defects are recorded separately.

## 8 Consequences and theorem boundary

The present result closes one precise gate and leaves several others untouched.

1. **Full-kernel parity simplicity is now proved.** At  $\sigma = 1/100$ , the negative continuum resonance required by the weighted-Riesz theorem is real, algebraically simple, and isolated.
2. **A continuum  $L^\infty$  resolvent is explicit.** The same circle carries the uniform bound 148.84153451786617.
3. **The exact continuum-normalized midpoint family is controlled.** Corollary 7.1 supplies an explicit all- $n$  statement from 32768 onward in the lifted  $L^\infty$  norm.

4. **Adaptive cutoff transfer is available in  $L^\infty$ .** Smooth discrete normalization converges at second order, while the adaptive support schedule of Wang [8] has vanishing row-operator defect. The resolvent perturbation theorem therefore transfers the isolated branch and its weighted Riesz term for all sufficiently fine adaptive grids in this norm.
5. **No uniform Euclidean sparse-matrix theorem is claimed.** The  $\ell^2$  condition required by the Euclidean cutoff proposition of Wang [13] is a different statement. Norm equivalence at a fixed dimension cannot be used uniformly as  $n \rightarrow \infty$ .
6. **The noise width is fixed.** All derivative constants grow as  $\sigma \downarrow 0$ . No small-noise or double-limit theorem follows from the present constants.

The center of the certified circle is a stored numerical datum, not a closed-form expression for  $\lambda_-$ . The theorem localizes the exact continuum eigenvalue only to the real interval cut out by the circle. It does not connect that eigenvalue to a zeta zero, construct a self-adjoint generator, prove a spectral counting law, or imply the Riemann hypothesis.

## 9 Archive and reproducibility

The repository archive separates three certificates:

1. the exact-stored 4096 Grushin contour certificate;
2. the 224-bit Arb stored-to-exact-midpoint bridge; and
3. the analytic Galerkin, Schur, and continuum composition certificate.

Each cross-paper input is recorded by path and SHA-256 hash. The final archive additionally hashes all local sources, publication artifacts, and the three result ledgers.

From the RH-41 directory, the complete replay is:

```
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
-m pytest -q -p no:cacheprovider
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/run_coarse_grushin_certificate.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/build_midpoint_bridge_certificate.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/build_continuum_certificate.py
MPLBACKEND=Agg PYTHONDONTWRITEBYTECODE=1 \
  /root/math/.venv/bin/python experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
cp main.pdf validated-parity-continuum-contour.pdf
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/build_archive.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/verify_archive.py
```

The Grushin replay is the expensive step because it validates the full 4097-dimensional bordered inverse in column chunks. The Arb midpoint bridge encloses all 4096 rows. The fast tests independently check the closed-form bounds and every archived theorem gate.

## 10 Conclusion

The negative peripheral branch has now crossed the continuum-isolation gate. A one-center Grushin problem converts exact stored residuals into a Rouché count, Arb bridges the stored binary64 matrix to the exact analytic midpoint kernel, exact-consistency Haar blocks propagate the contour through two refinements, and a final  $L^\infty$  perturbation closes the full continuum operator.

The outcome is an exact fixed-noise theorem: one algebraically simple real negative continuum resonance lies in the certified circle, and its contour resolvent is explicitly bounded. This supplies the missing premise for the full-kernel weighted-Riesz parity bridge and yields an all-fine-grid  $L^\infty$  midpoint statement. The route does not silently upgrade this bound to a Euclidean sparse-matrix theorem and does not pass to zero noise. Those are separate gates for later work.

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